Web Dev Mastery

🎨 CSS Typography and Styling

🎯 Topics Covered

- Fonts
- Font Properties
- Line Height
- Letter Spacing
- Text Decorations
- Colors (RGB, RGBA, HSL, HSLA)
- Background Color
- Background Image

1. Fonts

🔹 What is a Font?

A **font** is a particular style of text that determines how characters (letters, numbers, symbols) are displayed on the screen.

🔹 CSS Property: `font-family`

This defines which font to use. You can provide multiple fonts as a fallback list (browser uses the first one it can find).

🔸 HTML Example:

```
<p class="font-example">This is a font example.</p>
```

🔸 CSS Example:

```
.font-example {
  font-family: 'Arial', 'Helvetica', sans-serif;
}
```

If 'Arial' is not available, 'Helvetica' will be used. If none are available, the browser will use a default sans-serif font.

🔹 Custom Fonts (e.g., Google Fonts)

To use Google Fonts:

Add this to `<head>`:

```
<link
href="https://fonts.googleapis.com/css2?family=Roboto&display=swap"
rel="stylesheet">
```

CSS:

```
.font-example {
  font-family: 'Roboto', sans-serif;
}
```

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2. Font Properties

Font-related properties in CSS let you control the **style, weight, size, and variant** of your text.

🔹 Common Font Properties:

- `font-size`: Controls text size (e.g., `16px`, `1.2em`, `120%`)
- `font-weight`: Controls thickness (e.g., `normal`, `bold`, `100–900`)
- `font-style`: Sets italic or oblique
- `font-variant`: Enables small capital letters (`small-caps`)

🔸 HTML:

```
<p class="font-properties">Styled font text.</p>
```

🔸 CSS:

```
.font-properties {
  font-family: 'Georgia', serif;
  font-size: 20px;
  font-weight: bold;
  font-style: italic;
  font-variant: small-caps;
}
```

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3. Line Height

🔹 What is `line-height`?

Controls **vertical spacing between lines** in a paragraph or heading.

- Unitless value like `1.5` → relative to font size
- Fixed units like `24px` → absolute spacing

🔸 HTML:

```
<p class="line-height">
  This is a paragraph<br>
  with multiple lines<br>
  and adjusted line height.
</p>
```

🔸 CSS:

```
.line-height {
  font-size: 16px;
  line-height: 2; /* Multiplier of font size (2 × 16px = 32px) */
}
```

Why it matters?
Improves **readability** and avoids overlapping or cramped lines.

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4. Letter Spacing

🔹 What is `letter-spacing`?

Controls the **space between individual characters** (aka "tracking").

🔸 HTML:

```
<p class="letter-spacing">Letter Spacing Example</p>
```

🔸 CSS:

```
.letter-spacing {
  letter-spacing: 3px;
}
```

Use Cases:

- Logos
- Highlighted text
- Headings

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5. Text Decorations

🔹 What is `text-decoration`?

It is used to apply lines (underlines, strike-throughs, etc.) to text.

🔹 Values:

- `none` – no decoration
- `underline` – line below
- `overline` – line above
- `line-through` – line through text

🔸 HTML:

```
<p class="text-decoration">Text Decoration Example</p>
```

🔸 CSS:

```
.text-decoration {
  text-decoration: underline;
}
```

🔹 Advanced:

```
.text-decoration {
  text-decoration: underline overline red dashed;
}
```

This applies both underline and overline with red dashed lines.

6. Colors

🔹 `color` Property:

Used to change the color of text. You can define color in multiple formats:

🔹 Color Formats:

🟢 Named Colors:

```
color: red;
color: blue;
```

🟠 Hexadecimal (`#RRGGBB`):

```
color: #ff5733; /* Red-orange */
```

🔵 RGB – Red Green Blue:

```
color: rgb(255, 87, 51);
```

Full Form: `RGB = Red, Green, Blue`
Each value is from `0` to `255`.

🔴 RGBA – Red Green Blue Alpha:

```
color: rgba(255, 87, 51, 0.7);
```

A = Alpha (Transparency) from `0.0` (transparent) to `1.0` (opaque)

🟣 HSL – Hue Saturation Lightness:

```
color: hsl(14, 100%, 60%);
```

Full Form:

- **H**ue: angle (0–360°) on the color wheel
- **S**aturation: % intensity of color
- **L**ightness: % of light or brightness

7. Background Color

🔹 **`background-color` sets the color behind content.**

🔸 **HTML:**

```
<div class="bg-color">This div has a background color.</div>
```

🔸 **CSS:**

```
.bg-color {
  background-color: lightblue;
  color: black;
  padding: 10px;
}
```

Use Cases:

- Highlight sections
- Create themes

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8. Background Image

🔹 **`background-image` adds an image behind an element.**

🔸 **HTML:**

```
<div class="bg-image">This is a background image section.</div>
```

🔸 **CSS:**

```
.bg-image {
  background-image: url('https://via.placeholder.com/600x200');
  background-size: cover;          /* Fill area while maintaining
aspect ratio */
  background-repeat: no-repeat;    /* Prevent repeating */
  background-position: center;     /* Center the image */
  padding: 40px;
  color: white;
}
```

🔹 **Other Background Properties:**

- `background-repeat`: repeat, no-repeat, repeat-x, repeat-y
- `background-size`: auto, cover, contain, 100% 100%
- `background-position`: top, bottom, center, left, right
- `background-attachment`: scroll or fixed

✅ Summary Table

Property	Purpose
`font-family`	Font style
`font-size`	Size of the text
`font-weight`	Thickness of the text
`font-style`	Italic or normal style
`font-variant`	Small caps, etc.
`line-height`	Vertical spacing between lines
`letter-spacing`	Space between characters
`text-decoration`	Underline, line-through, etc.
`color`	Text color
`background-color`	Background behind the text
`background-image`	Image behind the element
`background-size`	Size of background image
`background-repeat`	Repeats image or not
`background-position`	Aligns the image within the element